FX Portfolio Lab:
From Portfolio Construction to FX Structuring

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Abstract

This working paper documents a research project that studies the foreign exchange (FX) market from three angles: the portfolio manager who allocates across currencies, the investor who hedges the currency risk of foreign assets, and the structurer who designs hedging products for corporate clients. We first build a valuation and backtesting engine whose correctness is verified against analytical results. We then construct G10 currency portfolios from carry, momentum and value signals and combine them with minimum-variance, mean-variance and risk-parity allocations, paying particular attention to estimation error and transaction costs. Next, we derive the optimal hedge ratio of a euro-based investor. Finally, we price and compare the structured products used to hedge FX exposures. Every design decision is recorded together with the alternatives considered and the reason for the choice (Appendix A).

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1 Introduction

The FX market is the largest financial market in the world, with an average daily turnover of about 7.5 trillion US dollars in April 2022 (Bank for International Settlements, 2022). Banks intervene in it in three main roles: they take and manage risk on behalf of investors, they help clients hedge their exposures, and they design structured solutions for corporate treasurers. This project reproduces, at a small scale and with public data, the analytical work behind each of these roles.

The project is organised around five research questions.

1. **Engine.** How can a currency portfolio be valued correctly, including forward positions, rebalancing and transaction costs? (Section 4)

2. **Currency portfolios.** Do the carry, momentum and value strategies documented in the literature (Lustig et al., 2011; Menkhoff et al., 2012a,b; Asness et al., 2013) survive realistic transaction costs, and which allocation method combines them best out of sample? (Section 5)

3. **The euro.** How much of the monthly moves of the euro is explained by euro-area rates and by sovereign risk? (Section 6)

4. **Hedging.** Which fraction of the currency exposure of foreign equities and bonds should a euro-based investor hedge to minimise risk? (Campbell et al., 2010) (Section 7)

5. **Structuring.** How do forwards, options and structured products (collars, participating forwards, target redemption forwards) compare for a corporate hedger, in terms of cost and distribution of outcomes? (Section 8)

The analytical framework of the first two questions is that of portfolio construction theory: valuation of rebalanced portfolios, first two moments of a portfolio return, minimum-variance and mean-variance optimisation (Markowitz, 1952; Milhau, 2026). A recurring theme is that the inputs of these optimisations are estimated, and that naive mean-variance portfolios are dominated out of sample by simple rules (DeMiguel et al., 2009).

Methodological principles. Three principles are applied throughout. (i) *No look-ahead bias*: every quantity used to form a portfolio at date t is computed from data available at t. (ii) *Verification*: every piece of code is checked against an analytical result or a published number before it is used (Appendix B). (iii) *Traceability*: every choice that could change the results is recorded in the decision log (Appendix A).

2 Framework and notation

2.1 Portfolio valuation

A portfolio invested in N constituents holds an amount $\theta_{t,i}$ in constituent i at date t. Its value and weights are

$$V_t = \sum_{i=1}^{N} \theta_{t,i}, \qquad w_{t,i} = \frac{\theta_{t,i}}{V_t}. \tag{1}$$

Between two rebalancing dates the portfolio is buy-and-hold: the number of units of each constituent is constant, so each amount grows with the constituent's own return,

$$\theta_{[t+1]^-,i} = \theta_{t,i}\,(1 + r_{t,t+1,i}), \tag{2}$$

and the portfolio return is the average of the constituents' returns weighted by the *initial* weights,

$$r_{t,t+1,p} = \sum_{i=1}^{N} w_{t,i}\, r_{t,t+1,i} = \boldsymbol{w}_t' \boldsymbol{r}_{t,t+1}. \tag{3}$$

On a rebalancing date the amounts are reset by the allocation rule; for a fixed-mix rule with target weights w_i, $\theta_{t+1,i} = w_i V_{t+1}$. The amount traded, $\sum_i |\theta_{t+1,i} - \theta_{[t+1]^-,i}|$, determines the transaction costs.

2.2 Moments and optimal portfolios

With $\boldsymbol{\mu}$ the vector of expected returns and $\boldsymbol{\Sigma}$ the covariance matrix of the constituents,

$$\mu_p = \boldsymbol{w}'\boldsymbol{\mu}, \qquad \sigma_p^2 = \boldsymbol{w}'\boldsymbol{\Sigma}\boldsymbol{w}. \tag{4}$$

The global minimum variance (GMV) portfolio and the maximum return-to-risk (MRR) portfolio are

$$\boldsymbol{w}_{\text{gmv}} = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}'\boldsymbol{\Sigma}^{-1}\mathbf{1}}, \qquad \boldsymbol{w}_{\text{mrr}} = \frac{\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}{\mathbf{1}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}}, \tag{5}$$

and, by the two-fund separation theorem, every efficient portfolio is a combination of the two (Milhau, 2026).

2.3 Currency excess returns

Let S_t be the spot exchange rate, in US dollars per unit of foreign currency, and i_t, i_t^* the domestic (USD) and foreign interest rates over $(t, t+1)$. A US investor who borrows one dollar at i_t and invests it in a foreign deposit earns the excess return

$$rx_{t+1} = (1+i_t^*)\frac{S_{t+1}}{S_t} - (1+i_t) \approx \underbrace{(i_t^* - i_t)}_{\text{carry}} + \underbrace{\frac{S_{t+1}-S_t}{S_t}}_{\text{spot change}}. \tag{6}$$

The same exposure is obtained with a forward contract. Under covered interest parity (CIP) the forward rate is $F_t = S_t(1+i_t)/(1+i_t^*)$, and a long forward position on $1/F_t$ units of foreign currency yields

$$\frac{S_{t+1}-F_t}{F_t} = \frac{rx_{t+1}}{1+i_t}. \tag{7}$$

The approximation in (6) shows the economic content of a currency position: the investor earns the interest rate differential and bears the risk of a depreciation of the foreign currency. Uncovered interest parity predicts that the two terms offset on average; the empirical failure of this prediction is the source of the carry trade premium (Lustig et al., 2011). Since 2008, CIP no longer holds exactly (Du et al., 2018); the consequences for our data are discussed in Section 3.

3 Data

All series are downloaded from two public sources, FRED (the database of the Federal Reserve Bank of St. Louis) and the ECB Data Portal, by the script `scripts/download_data.py` (Decision D7). Table 1 lists them.

Currency	Spot series	Spot available	Rate series	Rate available
USD	–	–	`IR3TIB01USM156N`	1964-06 – 2026-09
EUR	`DEXUSEU`	1999-01 – 2026-10	`IR3TIB01EZM156N`[a]	1994-01 – 2026-09
GBP	`DEXUSUK`	1971-01 – 2026-10	`IR3TIB01GBM156N`[b]	1957-01 – 2026-10
AUD	`DEXUSAL`	1971-01 – 2026-10	`IR3TIB01AUM156N`	1968-01 – 2026-08
NZD	`DEXUSNZ`	1971-01 – 2026-10	`IR3TIB01NZM156N`	1973-12 – 2026-09
JPY	`DEXJPUS`	1971-01 – 2026-10	`IR3TIB01JPM156N`	2002-04 – 2026-08
CHF	`DEXSZUS`	1971-01 – 2026-10	`IR3TIB01CHM156N`	1999-07 – 2026-09
CAD	`DEXCAUS`	1971-01 – 2026-10	`IR3TIB01CAM156N`	1956-01 – 2026-09
NOK	`DEXNOUS`	1971-01 – 2026-10	`IR3TIB01NOM156N`	1979-01 – 2026-09
SEK	`DEXSDUS`	1971-01 – 2026-10	`IR3TIB01SEM156N`	1982-01 – 2026-09

Table 1: Data series (downloaded on 20 October 2026). Spot: Federal Reserve H.10, daily noon buying rates in New York. Rates: OECD Main Economic Indicators, 3-month interbank rates, monthly averages, % per year. [a]Extended after January 2026 with the 3-month Euribor of the ECB (`FM.M.U2.EUR.RT.MM.EURIBOR3MD_.HSTA`). [b]Extended after January 2026 with the monthly average of SONIA (`IUDSOIA`), see Section 3.2.

Quotation convention. FRED quotes the euro, the pound and the Australian and New Zealand dollars in US dollars per unit of foreign currency, and the other currencies in foreign currency per US dollar. All spot rates are converted to US dollars per unit of foreign currency, so that an increase of S_t is always an appreciation of the foreign currency, consistently with equation (6).

Sample. The sample starts in January 1999, with the introduction of the euro. A currency enters the panel as soon as its spot rate and interest rate are both available (Decision D9): the Swiss franc in August 1999 and the yen in May 2002, all other currencies in February 1999. The euro and sterling rates published by the OECD stop in January 2026; they are extended to date (Section 3.2), so that all nine currencies remain in the panel until September 2026. A balanced panel (May 2002 – September 2026) is kept as a robustness check.

3.1 Descriptive statistics

Table 2 reports the monthly excess returns of a long position in each currency against the US dollar, computed as described in Section 4.2. The row EW is the equally weighted portfolio of the available currencies, rebalanced monthly; its return is the cross-sectional average of the excess returns. In the literature it is called the *dollar factor*, since it is short the US dollar against all other currencies (Lustig et al., 2011).

	First month	Last month	Months	Mean (%/yr)	Carry (%/yr)	Vol (%/yr)	Sharpe	Worst month (%)	Max DD (%)
EUR	1999-02	2026-09	331	-0.25	-0.60	9.16	-0.03	-9.78	42.41
GBP	1999-02	2026-09	331	0.15	0.49	8.46	0.02	-8.98	43.64
AUD	1999-02	2026-09	331	2.48	1.50	11.69	0.21	-16.53	38.78
NZD	1999-02	2026-09	331	2.75	1.81	12.58	0.22	-12.72	33.98
JPY	2002-05	2026-09	292	-2.06	-1.69	9.26	-0.22	-8.16	62.43
CHF	1999-08	2026-09	325	0.79	-1.80	9.71	0.08	-11.18	32.22
CAD	1999-02	2026-09	331	0.42	-0.11	8.24	0.05	-12.90	35.49
NOK	1999-02	2026-09	331	0.50	0.93	11.39	0.04	-11.94	53.62
SEK	1999-02	2026-09	331	-1.11	-0.66	10.86	-0.10	-10.62	53.15
EW	1999-02	2026-09	331	0.37	–	8.08	0.05	-8.89	39.16

Table 2: Monthly currency excess returns against the US dollar, February 1999 – September 2026. Mean, carry and volatility are annualised ($\times 12$ and $\times\sqrt{12}$). Carry is the average forward discount $S_t/F_t - 1$, the excess return earned if the spot rate does not move. Sharpe is the annualised mean divided by the annualised volatility (the returns are already in excess of the US rate). Max DD is the maximum drawdown. Generated by `scripts/describe_returns.py`.

Three facts emerge from Table 2 and Figure 1.

1. **Volatility.** Annualised volatilities lie between 8% and 13%, below those of an equity index (15–20%) but far from negligible: a currency position is a risky asset in its own right. The EW portfolio has a volatility of 8.1%, below that of every currency, as predicted by diversification.

2. **Carry and returns line up.** The currencies with the highest carry (AUD, NZD) have the highest average excess returns, and the yen, the currency with the lowest carry, has the lowest. This is the pattern exploited by the carry trade (Section 5). The Swiss franc is an exception: its carry is negative, yet its excess return is positive, because it appreciated against the dollar; this is consistent with its status as a safe-haven currency. Figure 1 also shows the asymmetry of carry returns: the cumulative value of the Australian dollar position fell by 32% between June 2008 and January 2009, more than ten years of its average excess return (2.5% a year) lost in seven months, which is the crash risk documented by Brunnermeier et al. (2009).

3. **Average returns are imprecise.** The t-statistic of a mean return is approximately the annualised Sharpe ratio times the square root of the number of years. With Sharpe ratios below 0.25 and about 27 years of data, no individual mean is significantly different from zero at the 5% level ($0.22 \times \sqrt{27.5} \approx 1.15 < 1.96$). Expected returns are therefore much harder to estimate than volatilities, which is the practical argument for the minimum-variance portfolio (Milhau, 2026).

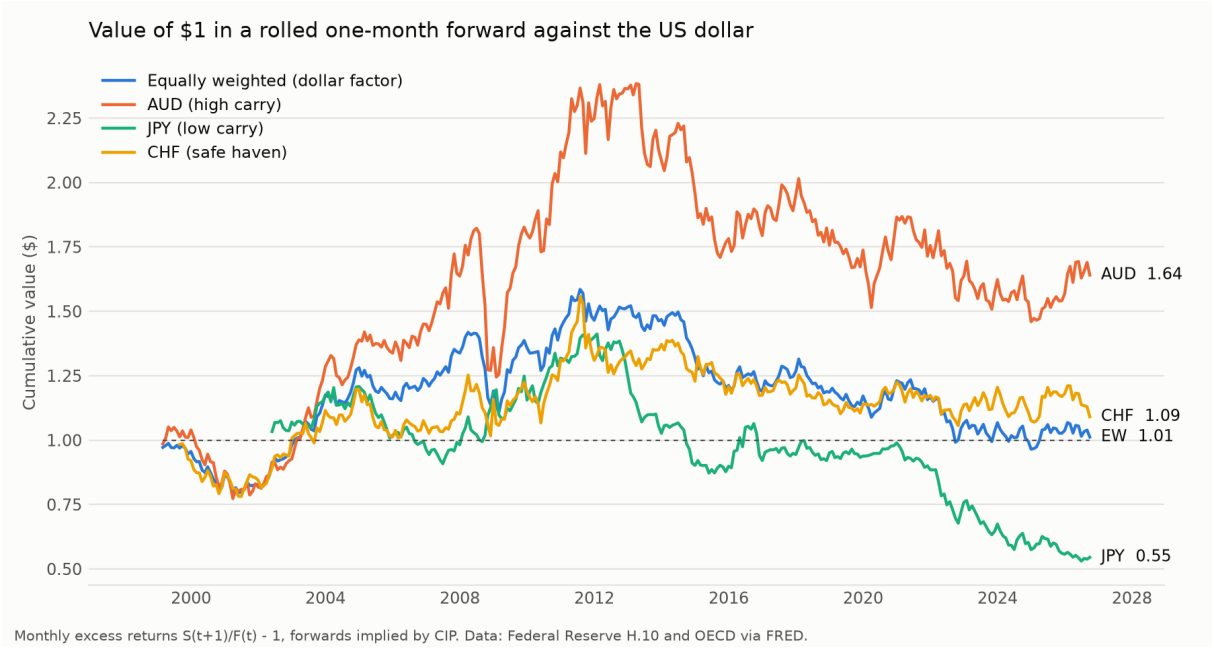


Figure 1: Cumulative value of $1 invested in a one-month forward rolled every month, for the equally weighted portfolio and three currencies with contrasting carry. Generated by `scripts/make_figures.py`.

3.2 Which interest rates in covered interest parity?

Interbank versus risk-free rates (Decision D16). The rates of the OECD are 3-month *interbank* rates: the rate at which a bank lends to another bank without collateral. They contain two components that the rate implicit in an FX forward does not necessarily contain. The first is a bank credit premium, since the loan is unsecured. The second is a term component: a 3-month rate reflects the policy rate expected over the next three months, so it rises above the overnight rate when the market expects hikes and falls below it when it expects cuts. The comparison of the sterling interbank rate with the overnight rate SONIA shows both: the gap was about +0.1 percentage point a year before 2021, rose to +0.6 in 2022 during the hiking cycle, and turned negative in 2024–2025 when cuts were expected. Since the reform of benchmark rates, FX forwards are priced off secured or overnight risk-free rates (SOFR, `ESTR`, SONIA), and an FX swap is itself a collateralised loan, of the same family as a repo. Before 2018–2019 these rates did not exist and the market priced forwards off interbank deposit rates, so the interbank rates are the consistent choice over most of the sample.

Robustness. Table 3 recomputes the euro and sterling excess returns from November 2019 with the monthly averages of the overnight risk-free rates instead of the interbank rates. The average carry changes by at most 0.03 percentage point a year, the monthly excess returns differ by about 1 basis point on average, and their correlation is above 0.9999. The choice of rate is therefore immaterial for the results of this paper; it would matter for the pricing of an individual forward, where 1 basis point a month is the order of magnitude of a bid–ask spread.

Extension after January 2026 (Decision D17). The OECD series for the euro is the 3-month Euribor: it coincides with the ECB series to within 0.0001 percentage point on average over 1994–2026, so the ECB series simply continues it. For sterling, no 3-month interbank rate exists after the end of LIBOR; the series is continued with the monthly average of SONIA, shifted by the average gap between the two series over the last twelve months of overlap (−0.08

	Months	Carry (%/yr)		Mean rx (%/yr)		Mean $\lvert\Delta rx\rvert$	Corr.
		Interbank	RFR	Interbank	RFR	(bp/month)	
EUR	74	-1.49	-1.46	-0.45	-0.42	1.0	1.0000
GBP	74	-0.17	-0.19	1.47	1.46	0.9	1.0000

Table 3: Excess returns against the US dollar with 3-month interbank rates (baseline) and with monthly averages of the overnight risk-free rates (SOFR for USD, `ESTR` for EUR, SONIA for GBP), November 2019 – January 2026. Generated by `scripts/robustness_rfr.py`.

percentage point), so that the spliced series has no jump.

3.3 Limitations

- **Forward rates** are not freely available; they are implied by CIP (equation 7). Since 2008, CIP deviations of the order of a few tens of basis points per year exist (Du et al., 2018).
- **Interest rates** are monthly averages of interbank rates, not the rate observed on the day a position is opened; the choice of interbank rather than risk-free rates is discussed in Section 3.2.
- **Maturity mismatch**: 3-month rates are used to price 1-month forwards, because 1-month rates are not freely available for all currencies. The error on the carry is of the order of the slope of the money-market curve between 1 and 3 months.
- **Transaction costs**: bid–ask spreads are not in the data; they are modelled as a stated assumption with a sensitivity analysis (Section 4.4).

4 Valuation and backtesting engine

4.1 Portfolio valuation

The engine values a portfolio by propagating the dollar amounts held in each constituent, following the general method of Milhau (2026): amounts are updated with equation (2) between rebalancing dates and reset by the allocation rule on rebalancing dates. Working with amounts rather than weights has two advantages. First, it handles any rebalancing frequency with the same code (daily rebalancing and buy-and-hold are the two limiting cases). Second, it gives the amount traded at each date directly, from which transaction costs are computed (Decision D4).

Verification. The engine reproduces the worked examples of the course: the valuation of an equally weighted portfolio of five US stocks between 2 and 5 January 2015 (final value \$978.21, \$6.37 traded to return to equal weights), the drift of weights in a buy-and-hold portfolio, and the compounding of returns. These examples are encoded as automated tests (Appendix B).

4.2 Monthly currency excess returns

Sampling. The month-end spot rate is the last available daily rate of the month (Decision D8); if the data stop before the last business day of the final month, that month is dropped, so that no mid-month rate is treated as a month-end rate. The OECD interest rate of month t is an average over the month, so it is dated at the end of month t, when it is fully known.

Forward rate and excess return. At the end of month t, the one-month forward rate implied by CIP is

$$F_t = S_t \frac{1 + i_t/12}{1 + i_t^*/12}, \tag{8}$$

and the excess return of a long forward position held over month $t+1$ is (Decision D11)

$$rx_{t+1} = \frac{S_{t+1}}{F_t} - 1. \tag{9}$$

The *carry* of the position, known at t, is $S_t/F_t - 1 \approx (i_t^* - i_t)/12$: the return earned if the spot rate does not move.

Timing. Equation (9) uses only S_t, i_t and i_t^*, observed at the end of month t, to open the position; the outcome S_{t+1} is observed one month later. A test checks that changing the interest rates at $t+1$ or the spot rates after $t+1$ leaves rx_{t+1} unchanged.

4.3 Backtesting portfolios of forwards

The valuation of Section 4.1 assumes a fully invested portfolio, whose weights sum to one. A currency strategy is different: a forward requires no capital, so the capital V_t is held as collateral, and the strategy is described by *notional weights* $w_{t,i} = N_{t,i}/V_t$, the notional of the forward on currency i divided by the capital. These weights need not sum to one; a long-short strategy has $\sum_i w_{t,i} = 0$.

The engine applies the amount-propagation method to notionals (Decision D4). At the end of month t, the strategy decides target weights w_t from information available at t, and the notionals are set to $N_{t,i} = w_{t,i}V_t$. Moving from the notionals of the previous month to N_t costs C_t (Section 4.4), paid immediately. Over month $t+1$ the forwards earn $\sum_i N_{t,i}\, rx_{t+1,i}$ and the collateral may earn the US rate, so that

$$V_{t+1} = V_t - C_t + \sum_{i=1}^{N} N_{t,i}\, rx_{t+1,i} + V_t \frac{i_t}{12}, \qquad \frac{V_{t+1}}{V_t} - 1 = \boldsymbol{w}_t' \boldsymbol{rx}_{t+1} - \frac{C_t}{V_t} + \frac{i_t}{12}. \tag{10}$$

The first term is the course's fundamental property $r_p = \boldsymbol{w}'\boldsymbol{r}$ applied to excess returns; the last term is omitted when excess returns are reported, which is the default in this paper. A position in a currency whose excess return is not available is rejected by the engine, so that a strategy cannot silently trade a currency that has left the panel.

4.4 Transaction costs

Model (Decision D14). A forward position kept from one month to the next is rolled with an FX swap, which simultaneously closes the maturing forward and opens a new one; a swap is cheaper than two outright trades. Any change of position is an outright trade. For each currency, with N^- the previous notional and N the new one, the cost is

$$C_i = c_i^{\text{swap}} \times \underbrace{\min(|N^-|, |N|)\mathbb{1}\{N^- N > 0\}}_{\text{notional rolled}} + c_i^{\text{out}} \times \underbrace{|N - N^-|}_{\text{notional traded}}, \tag{11}$$

where c_i^{swap} and c_i^{out} are half-spreads in fraction of notional. A change of sign keeps nothing: the position is closed and reopened. A model that charged only the notional traded would ignore the roll and understate the cost of strategies that hold their positions, such as the carry trade.

Level (Decision D15). Bid–ask spreads are not freely available. The base case is therefore an *assumption* by liquidity tier, not an estimate: outright and swap half-spreads of 2 and 1 basis points for the most liquid currencies (EUR, JPY, GBP, CHF, CAD), 3 and 1.5 for AUD and NZD, and 5 and 2.5 for NOK and SEK. Rather than depending on these figures, every strategy is reported for the base case scaled by 0, 1, 2 and 5, together with its *break-even cost scale*: the multiple of the base case at which its average net return falls to zero. Readers can compare this multiple with their own trading costs, which differ widely between an interbank dealer and a corporate client.

4.5 Performance and risk measures

For monthly returns r_t with sample mean $\bar{r}$ and standard deviation s over n months, the paper reports the annualised mean $12\bar{r}$, the annualised volatility $\sqrt{12}\,s$, the Sharpe ratio $\sqrt{12}\,\bar{r}/s$ (the returns being already in excess of the US rate), the t-statistic of the mean $\bar{r}/(s/\sqrt{n})$, which equals the Sharpe ratio times the square root of the number of years, the skewness and excess kurtosis of monthly returns, the maximum drawdown, the annual cost $12\,\overline{C/V}$ and the annual turnover $12\,\overline{\sum_i |N_t - N_t^-|/V}$. Skewness is the key measure of crash risk: a strategy that earns small regular gains and suffers rare large losses has a negative skewness (Brunnermeier et al., 2009).

4.6 First application: the dollar factor

The equally weighted portfolio of Table 2 is now backtested with the engine: at the end of each month it is long $1/N_t$ of each of the N_t currencies available at that date, i.e. whose spot rate and interest rates are known. Table 4 reports the results for the four cost scenarios.

Costs	Mean (%/yr)	Vol (%/yr)	Sharpe	t-stat	Skew.	Max DD (%)	Costs (%/yr)	Turnover (×/yr)
0 × base case	0.37	8.07	0.05	0.24	-0.04	39.2	0.000	0.34
1 × base case	0.19	8.07	0.02	0.12	-0.04	40.6	0.181	0.34
2 × base case	0.01	8.07	0.00	0.00	-0.04	42.0	0.362	0.34
5 × base case	-0.54	8.07	-0.07	-0.35	-0.04	46.1	0.905	0.34

Table 4: Equally weighted currency portfolio (dollar factor), February 1999 – September 2026, monthly excess returns net of transaction costs. Costs are the base case of Section 4.4 multiplied by 0, 1, 2 and 5. Generated by `scripts/backtest_benchmarks.py`.

Three points are worth noting. First, with zero costs the engine reproduces the mean of the cross-sectional average of Table 2 (0.37% a year), an independent check of the backtest. Second, even a strategy that almost never trades pays about 0.18% a year in the base case, because its whole notional is rolled every month; the outright turnover is only 0.34 times the capital a year, coming from the monthly resizing of positions and from currencies entering the panel. This confirms that ignoring the roll (alternative of Decision D14) would understate costs. Third, the break-even cost scale is 2.0 times the base case: with costs about twice the base case, the small premium of the dollar factor disappears. The dollar factor is not a strategy in itself but the benchmark against which the long-short strategies of Section 5 are compared; since those are long and short at the same time, they do not earn this premium and must be justified by their own returns.

5 Currency portfolios

[TODO: Phase 2.]

- Signals: carry (interest rate differential), momentum (past excess returns), value (deviation from purchasing power parity).
- Portfolio formation: sorting into long-short portfolios versus optimisation.
- Allocation across strategies: equal weights, GMV, mean-variance, risk parity; covariance estimation with shrinkage (Ledoit and Wolf, 2004).
- Crash risk of the carry trade (Brunnermeier et al., 2009).

6 What moves the euro: the rates channel and the risk channel

On a G10 FX desk, the euro is driven first by rates. Two channels link euro-area rates to the exchange rate. In the *rates channel*, a rise of euro rates relative to foreign rates makes euro assets more attractive and raises the euro; it also changes the forward points through covered interest parity. In the *risk channel*, doubts about the sustainability of the debt of a member state widen its sovereign spread over the German Bund, a risk premium appears on the euro and the euro falls, especially against the Swiss franc, the European safe haven. This section measures both channels with public data. It prepares the hedging problem of Section 7: a euro-based investor needs to know when the euro is likely to fall.

6.1 Specification

The variables are monthly changes of monthly averages (Decision D18). The dependent variable is the percentage change of the monthly average of EURUSD or EURCHF, positive when the euro appreciates. The rates channel is measured by the change in the 3-month rate differential, $\Delta(i_{\text{EUR}} - i_{\text{USD}})$ or $\Delta(i_{\text{EUR}} - i_{\text{CHF}})$, and by the change in the 10-year differential between the Bund and the US Treasury. The risk channel is measured by the change in the 10-year spreads of Italy (BTP) and France (OAT) over the Bund. Rates, spreads and their changes are in percentage points. Each regression

$$\Delta s_t = \alpha + \beta_{\text{rates}}\,\Delta(\text{rate differential})_t + \beta_{\text{risk}}\,\Delta(\text{spread})_t + \varepsilon_t \tag{12}$$

is estimated by ordinary least squares with Newey–West standard errors (Newey and West, 1987), which remain valid when the residuals are heteroskedastic or autocorrelated; the latter is expected here because averaging daily data over a month induces autocorrelation in monthly changes. The coefficients are contemporaneous associations, not causal effects: both sides react to the same news within the month.

6.2 Results

The risk channel is significant against both currencies. In column (1), a widening of the BTP–Bund spread by 10 basis points is associated with a fall of the euro against the dollar of about 0.23% in the same month ($t = -4.1$). The effect survives the addition of the 10-year rate differential and of the French spread in column (2). Against the Swiss franc, the full-sample estimate in column (3) is small and not significant, but this is an artefact of the Swiss National Bank floor: from September 2011 to January 2015 the franc was capped at 1.20 per euro, precisely during the euro-area debt crisis, so EURCHF could not respond to euro risk. Without these months, in column (4), the coefficient is -2.33 ($t = -3.3$), the same order of magnitude as against the dollar. The negative constant of the EURCHF regressions, -0.15% a month, is the trend appreciation of the franc against the euro over the sample, about 1.8% a year.

	(1) EURUSD	(2) EURUSD	(3) EURCHF	(4) EURCHF ex-floor
$\Delta(i_{\text{EUR}} - i_{\text{USD}})$, 3M	1.08	0.20		
	(1.1)	(0.2)		
$\Delta(i_{\text{EUR}} - i_{\text{CHF}})$, 3M			-1.00	-0.33
			(-1.0)	(-0.5)
$\Delta(y_{\text{Bund}} - y_{\text{UST}})$, 10Y		5.76		
		(5.1)		
Δ BTP–Bund, 10Y	-2.27	-2.65	-0.65	-2.33
	(-4.1)	(-4.3)	(-1.0)	(-3.3)
Δ OAT–Bund, 10Y		1.07		
		(0.5)		
Constant	0.03	0.04	-0.15	-0.16
	(0.2)	(0.3)	(-2.1)	(-2.1)
R^2	0.05	0.16	0.02	0.09
Months	329	329	325	284

Table 5: Regressions of the monthly percentage change of the euro on the changes in rate differentials and sovereign spreads (percentage points), February 1999 – August 2026. t-statistics with Newey–West standard errors (5 lags) in parentheses. Column (4) excludes the period of the Swiss National Bank floor on EURCHF (September 2011 – January 2015). Generated by `scripts/euro_drivers.py`.

The rates channel works through long rates. The change in the 3-month differential is not significant in any column, whereas the change in the 10-year differential between the Bund and the US Treasury is the strongest variable of column (2): a 10-basis-point rise of the Bund relative to the Treasury goes with a rise of the euro of about 0.58% ($t = 5.1$). Two reasons explain the difference. First, short rates move with policy decisions, which are largely anticipated; the exchange rate reacts to changes in the expected path of policy, which long rates capture. Second, monthly averages of short rates are smooth, so their changes contain little news. The French spread adds no information once the Italian spread is in the regression.

The risk channel is not constant. Figure 2 reports the coefficient on the BTP–Bund spread over rolling 36-month windows. Before 2008 it is erratic, because sovereign spreads were a few tens of basis points and barely moved, so the coefficient is poorly identified. Against the dollar it reached about -1% per 10 basis points at the height of the debt crisis in 2011, and it has been negative since 2018, between about -0.1% and -0.7%; against the franc it has been of similar size since 2018. In between, for windows ending in 2016–2018, it turned positive, a period dominated by the asset purchases of the ECB, which lowered both the euro and Italian spreads; this is consistent with the two channels pulling in opposite directions when the central bank itself drives the spread.

6.3 What is left out, and why

A trader would also follow ECB meetings, sovereign debt auctions and the carry and roll-down of bonds. These are monitoring and rates-desk topics rather than questions that public data can answer for 27 years; they are left out of the project, for the reasons given in Decision D19.

7 Optimal currency hedging for a euro-based investor

[TODO: Phase 3.]

The minimum-variance hedge ratio is a direct application of the two-asset minimum-variance weight $w_2^* = (\sigma_1^2 - \sigma_{12})/\sigma_{2/1}^2$ (Milhau, 2026), where the two assets are the unhedged foreign asset

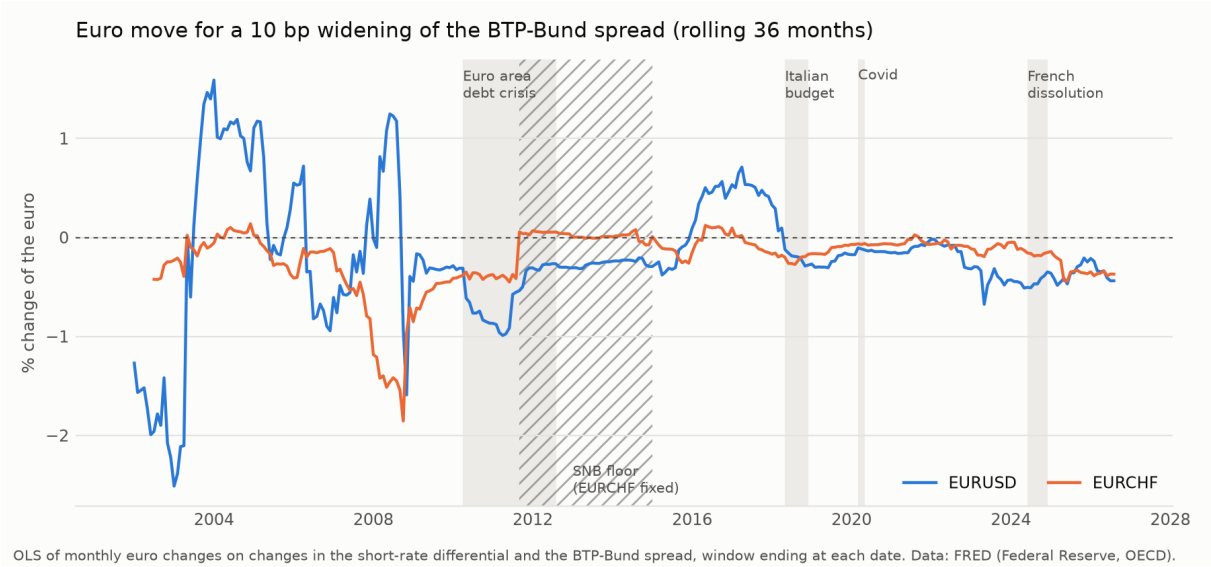


Figure 2: Coefficient on the change in the BTP–Bund spread in regressions of the monthly change of EURUSD and EURCHF on the short-rate differential and the spread, over rolling 36-month windows ending at each date, scaled to a 10-basis-point widening. Hatched: Swiss National Bank floor. Generated by `scripts/euro_drivers.py`.

and the currency forward. The empirical question is whether some currencies, such as the US dollar, act as natural hedges for equity risk (Campbell et al., 2010).

8 FX structuring

[TODO: Phase 4.]

- Option pricing: Garman and Kohlhagen (1983); volatility smile from at-the-money, risk-reversal and butterfly quotes.
- Products: forward, vanilla option, collar, participating forward, target redemption forward (Monte Carlo).
- Comparison for a corporate hedger: upfront cost, distribution of the hedged rate, worst-case outcomes.

9 Conclusion and limitations

[TODO: Written at the end of the project.]

Known limitations at this stage:

- Forward rates are implied by covered interest parity rather than observed.
- Transaction costs are modelled, not observed.

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A Decision log

Each entry records a choice that could affect the results or the usability of the project, the alternatives that were considered and the reason for the choice. Entries are never deleted: a decision that is reversed is superseded by a new entry that refers to it.

D1 – Language of the project

Date 18 October 2026

Decision The paper, the code and its comments are written in English.

Alternatives French.

Rationale English is the working language of trading floors and of open-source code, and the project is meant to be shown to recruiters, including outside France.

D2 – Code organised as an installable Python package

Date 18 October 2026

Decision The code lives in a package (`src/fxlab`) installed in a virtual environment; notebooks only import and use it.

Alternatives (a) Code written directly in Jupyter notebooks. (b) Loose scripts.

Rationale Code in notebooks cannot be tested or reused, and is easily run out of order. A package separates the library (tested, reused by every phase) from the analysis (notebooks). The `src` layout guarantees that the tests run against the installed package, not against stray files.

D3 – Course examples used as automated tests

Date 18 October 2026

Decision Every worked example of the course that the code can reproduce is encoded as a test (`pytest`), with a tolerance that reflects the rounding of the published inputs.

Alternatives Checking results by eye in notebooks.

Rationale The course examples have known answers, so they are an independent specification of what the code must do. Automated tests are rerun after every change, which prevents an improvement in one place from silently breaking another.

D4 – Valuation by propagation of amounts

Date 18 October 2026

Decision Portfolios are valued by propagating the amount held in each constituent and resetting it on rebalancing dates (course, section 1.8).

Alternatives Compounding the portfolio return $\boldsymbol{w}'\boldsymbol{r}$ with fixed weights.

Rationale Compounding with fixed weights is only valid when the portfolio is rebalanced at every date. Propagating amounts handles any rebalancing frequency and gives the amount traded, which is needed for transaction costs.

D5 – Free, official data sources

Date 18 October 2026

Decision Exchange rates and interest rates are taken from official publishers (Federal Reserve, OECD, ECB) through FRED and the ECB data portal; forward rates are implied by CIP.

Alternatives (a) Bloomberg or Refinitiv: no access. (b) Yahoo Finance: unofficial interface, no documentation of the series, known data errors.

Rationale Official sources are documented, stable and allow anyone to reproduce the results. The cost is that forward rates are not observed; this is acknowledged as a limitation (Section 3).

D6 – Working paper in LaTeX, versioned with the code

Date 18 October 2026

Decision The documentation is a LaTeX article stored in the same repository as the code.

Alternatives (a) A word processor document. (b) Notebooks only.

Rationale LaTeX is the standard for scientific documents with equations. Keeping it in the repository ties each version of the text to the version of the code that produced its results.

D7 – Data downloaded by a script from the FRED CSV interface

Date 19 October 2026

Decision A script downloads each series from its public FRED address and caches the raw file in `data/raw/`; the files are read and cleaned by the `fxlab.data` module. Spot rates are converted to US dollars per unit of foreign currency.

Alternatives (a) Manual download from the website. (b) The FRED API, which requires a personal key. (c) A third-party package such as `pandas-datareader`.

Rationale A script makes the dataset reproducible with one command and can refresh it at any time. The public CSV interface needs no key and only the Python standard library, so there is no dependency that could break. A single quotation convention avoids sign errors on half of the currencies.

D8 – Monthly frequency, month-end sampling

Date 19 October 2026

Decision Currency strategies are formed and evaluated monthly, using the last available spot rate of each month.

Alternatives (a) Daily frequency. (b) Weekly frequency. (c) Monthly averages of spot rates.

Rationale Interest rates are only available monthly, so a higher frequency would reuse the same rate for a whole month. Monthly rebalancing is the standard of the currency literature (Lustig et al., 2011; Menkhoff et al., 2012a), which makes the results comparable, and it keeps turnover and transaction costs low. Month-end rates, not averages, are used because a position is opened and closed at a given rate, not at an average.

D9 – Unbalanced panel from January 1999

Date 19 October 2026

Decision The sample starts in January 1999; each currency enters as soon as its spot rate and interest rate are both available. A balanced panel (2002–2026) is used as a robustness check.

Alternatives (a) Balanced panel from May 2002 only. (b) Start in the 1980s without the euro.

Rationale Starting with the euro gives a homogeneous G10 universe. An unbalanced panel uses three more years of data without inventing any, as in Lustig et al. (2011); the robustness check verifies that the conclusions do not depend on this choice.

D10 – US dollar as base currency for the currency portfolios

Date 19 October 2026

Decision The excess returns of Phase 2 are measured from the point of view of a US investor; the euro-based investor is the subject of Phase 3.

Alternatives Euro as base currency from the start.

Rationale The data are quoted against the US dollar and almost all of the literature uses a US investor, so the results can be compared with published ones. Excess returns of long-short portfolios do not depend much on the base currency, since the base currency cancels out of a long-short position.

D11 – Excess return defined as the simple return of a one-month forward

Date 19 October 2026

Decision $rx_{t+1} = S_{t+1}/F_t - 1$ with F_t implied by CIP and the monthly rate equal to the annual rate divided by 12 (equations 8–9).

Alternatives (a) Log excess return $\ln S_{t+1} - \ln F_t$, common in the literature. (b) Deposit version (6).

Rationale The portfolio formula $r_p = \boldsymbol{w}'\boldsymbol{r}$ of the course holds for simple returns, not for log returns, so simple returns can be aggregated across currencies without approximation. The forward is the instrument actually traded by an FX desk for this exposure, and it requires no capital beyond margin. The deposit version differs only by the factor $1 + i_t$ (equation 7).

D12 – Continuous integration and MIT licence

Date 19 October 2026

Decision The test suite runs automatically on GitHub Actions at every push, on Python 3.11 and 3.13; the code is published under the MIT licence.

Alternatives (a) Running the tests locally only. (b) No licence, or a restrictive one.

Rationale Tests run on a clean machine catch errors that a local environment hides (a missing dependency, a file that only exists locally), and the badge shows any reader that the suite passes. Without a licence, published code cannot legally be reused; MIT is the most common licence for demonstration projects and imposes only attribution.

D13 – Compiled paper versioned in the repository

Date 19 October 2026

Decision The compiled PDF is copied to `paper/fx-portfolio-lab.pdf` by `make` and committed with each update of the paper; LaTeX build files remain ignored.

Alternatives (a) PDF attached to GitHub releases only. (b) PDF compiled by the continuous integration. (c) Sources only.

Rationale A reader who is not going to compile LaTeX must be able to open the paper in one click from the README. Committing the PDF is the simplest way to guarantee it and keeps each version of the PDF next to the code that produced it. Compiling in the CI would require installing a TeX distribution on every run, which is slow for a document that changes only a few times per phase.

D14 – Transaction costs: roll plus change of position

Date 19 October 2026

Decision The notional kept from one month to the next pays the swap half-spread (roll); the notional traded pays the outright half-spread (equation 11).

Alternatives A single half-spread on the notional traded only.

Rationale Holding a forward position requires rolling it every month with an FX swap, which is how an FX desk maintains the exposure. Ignoring the roll would understate the cost of strategies that hold their positions; on the dollar factor, the roll accounts for most of the cost (Section 4.6).

D15 – Cost level: stated assumption, sensitivity and break-even

Date 19 October 2026

Decision The base-case half-spreads are an explicit assumption by liquidity tier (Section 4.4); results are reported for the base case scaled by 0, 1, 2 and 5, with the break-even cost scale of each strategy.

Alternatives (a) Figures presented as estimates without a public data source. (b) A single cost for all currencies.

Rationale Bid–ask quotes are not freely available, so any single figure would be an assumption; presenting it as an estimate would overstate its precision. The break-even scale answers the question that matters (up to which cost does a strategy remain profitable?) without depending on the assumption, and the tiers keep the ranking of liquidity between currencies.

D16 – Interbank rates in covered interest parity, with a risk-free-rate robustness check

Date 20 October 2026

Decision Forwards are implied from 3-month interbank rates over the whole sample; from November 2019, euro and sterling excess returns are recomputed with the overnight risk-free rates (SOFR, ESTR, SONIA) as a robustness check (Section 3.2).

Alternatives (a) Risk-free rates only: they do not exist before 2018–2019 and not for every currency in free data. (b) Interbank rates without discussion.

Rationale Interbank rates contain a bank credit premium and a term component that the rates of a collateralised FX swap do not contain, a point raised by practitioners. But they were the market reference before the benchmark reform, they cover the whole sample and all currencies, and the robustness check shows that switching to risk-free rates changes monthly excess returns by about 1 basis point. The choice is therefore kept and its limits are stated.

D17 – Extension of the euro and sterling rates after January 2026

Date 20 October 2026

Decision The euro rate is continued with the 3-month Euribor published by the ECB; the sterling rate with the monthly average of SONIA, shifted by the mean gap over the last twelve months of overlap.

Alternatives (a) Let the two currencies leave the panel in February 2026. (b) Splice SONIA without a shift.

Rationale The OECD euro series is the Euribor, so the ECB series continues it exactly. For sterling no 3-month interbank rate exists after LIBOR; a constant shift removes the jump at the junction (-0.08 point) without changing the monthly variation of the rate. Keeping the euro and sterling in the panel avoids changing the investment universe at the end of the sample.

D18 – Euro drivers measured with monthly averages and Newey–West standard errors

Date 20 October 2026

Decision The regressions of Section 6 use changes of monthly averages for the exchange rates as well as for rates and yields, and Newey–West standard errors with the usual lag rule.

Alternatives (a) Month-end exchange rates against monthly-average yields. (b) Classical OLS standard errors.

Rationale The OECD rates and yields are only available as monthly averages; mixing them with month-end exchange rates would misalign the timing of the two sides. Averaging induces autocorrelation in monthly changes, and financial data are heteroskedastic, so classical standard errors would overstate precision; Newey–West standard errors are robust to both.

D19 – Scope: what is not added from the practitioner advice

Date 20 October 2026

Decision ECB meeting pricing, sovereign debt auctions and the carry and roll-down of bonds are not part of the project; they are followed as market monitoring outside it.

Alternatives (a) An event study of ECB meeting days. (b) A rates module for bond carry, roll-down and repo.

Rationale Measuring a monetary policy surprise requires the market pricing of each meeting (dated overnight-index swaps), which is not freely available historically, and a complete list of meeting dates that would have to be compiled by hand; auctions have no free historical database. Bond carry and roll-down belong to a rates desk: adding them would blur a project whose subject is currencies. The euro is studied instead through the two channels of Section 6, which public data can measure.

Open decisions

None at this stage. The decisions of Phase 2 (signal construction, portfolio formation, estimation of the covariance matrix) will be added before the corresponding code is written.

B Software architecture and verification

Module	Content
`fxlab.portfolio`	Returns, compounding, valuation of buy-and-hold and rebalanced portfolios
`fxlab.data`	Download and loading of FRED and ECB series, quotation convention, splicing
`fxlab.fx_returns`	Month-end sampling, CIP forwards, excess returns, carry
`fxlab.costs`	Cost model: roll and outright half-spreads, base case and scaling
`fxlab.forwards`	Backtest of portfolios of rolled forwards, equal weights
`fxlab.metrics`	Performance and risk measures, break-even cost scale
`fxlab.regression`	OLS with Newey–West standard errors, rolling coefficients
`fxlab.euro`	Monthly changes of the euro and of its rates and risk drivers

Table 6: Modules of the `fxlab` package.

Test	Source	Result checked
`test_table1_buy_and_hold`	Course, Table 1	$V_1 = \$978.21$, return -2.18%
`test_table1_rebalancing_...`	Course, section 1.4	\$6.37 traded
`test_midterm_q1_amount_update`	Midterm 2024–25, Q1	\$36.29
`test_midterm_q4_weight_drift`	Midterm 2024–25, Q4	weight 74.9%
`test_midterm_q7_compounding`	Midterm 2024–25, Q7	+0.8%
`test_single_rebalance_...`	Course, section 1.8	equals buy-and-hold
`test_spot_converted_...`	FRED conventions	JPY per USD inverted
`test_missing_values_...`	FRED format	missing values are NaN
`test_rates_in_decimal`	FRED format	5% read as 0.05
`test_forward_by_cip`	Eq. (8)	forward computed by hand
`test_excess_return_by_hand`	Eq. (9)	excess return computed by hand
`test_equal_rates_...`	Eq. (9)	no carry: rx = spot return
`test_carry_...`	Section 4.2	carry $\approx (i^* - i)/12$
`test_no_look_ahead`	Timing rule	rx_{t+1} unaffected by later data
`test_month_end_spot_...`	Sampling rule	last available day used
`test_incomplete_last_month_...`	Sampling rule	partial month dropped
`test_rates_dated_...`	Timing rule	rate dated at month end
`test_costs.py` (10 tests)	Eq. (11)	opening, roll, resize, sign change, scaling, by hand
`test_single_long_...`	Eq. (10)	one forward earns rx
`test_long_short_...`	Eq. (10)	weights summing to 0: $r_p = \boldsymbol{w}'\boldsymbol{rx}$
`test_collateral_...`	Eq. (10)	collateral earns $i_t/12$
`test_costs_by_hand`	Eqs. (10)–(11)	costs of two months by hand
`test_turnover_of_a_flip`	Section 4.5	long to short trades twice the notional
`test_no_look_ahead` (forwards)	Timing rule	weights at t only affect month $t+1$
`test_position_without_...`	Section 4.3	position without a return rejected
`test_metrics.py` (7 tests)	Section 4.5	mean, vol, Sharpe, t-stat, drawdown, skewness by hand
`test_data.py`, new tests (5)	Section 3.2	ECB file format, monthly averages, splice without jump, overlap required
`test_regression.py` (8 tests)	Section 6	coefficients recovered, White formula, HAC wider under autocorrelation, lag rule, rolling, drivers by hand

Table 7: Verification tests.